MechToolBox

Target Applications

Research projects that employ *in situ* loading HEXD experiments (both near-field and far-field configurations) to probe structure-property-response relations at the crystal scale.

Range of Capabilities

Finite element-based analyses of the mechanical behaviors of virtual samples, including:

- Linear elastic analysis of mechanical loading of virtual samples typically performed in experiments (uniaxial and biaxial extension and compression)
- Evaluation of crystal-based mechanical metrics related to stiffness, strength and their ratios.
- Evaluation of elastic moduli using measured grain-averaged elastic strain tensors.
- Reconstruction of equilibrated stress distributions using measured grain-averaged elastic strain tensors.
- Modal analyses of field variables, including stress-related and deformation-related fields.
- Connected set analyses to examine banding of structure, deformation and stress fields.

Code Attributes

MatLab-based scripts residing in a GitHub repository. Interfaces with Neper for instantiation of virtual samples and with Paraview/Visit for visualization.

Incorporation of MechToolBox Capabilities in a HEXD Project Workflow



Obtained via EBSD and/or HEDM and analyzed via DREAM3D and Ormonics.

Performed with Neper (or alternative code)

Performed with:
MechMet (elastic responses)
or
FEpX (elastoplastic responses)

Performed with MechVDE

Several possibilities to analyze data, including MechMonics, MechModuli, MechBands, and MechRecon.

MechToolBox codes assist with pre-test planning, test set-up, and post-test interpretations:

In Pre-Test Planning the goal is to build hypotheses regarding possible correlations of behavior and structure. Analyses can include predicting stress and strain distributions, assessing the effects of state and properties, and identifying morphological and topological dependences.

For Test Set-up the goals are to determine if an HEXD experiment is well-suited to testing a hypothesis, and if so, to propose configurations for the beam, detector, and loading to most effectively collect desired experimental data.

In Post-Test Interpretations, the goals are many and varied, including assessing the hypothesis, estimating properties related to strength and stiffness, and identifying structure-property correlations.

Codes: Green coloring designates MechToolBox codes
Purple coloring designates a sister code in a separate repository

MechToolBox

Main Codes in MechToolBox:

- **MechMet**: Performs linear elastic analyses of virtual polycrystal and evaluates mechanical (elastoplastic) metrics.
- **MechMonics**: Computes discrete harmonic functions for grains within a virtual polycrystal.
- **MechReCon**: Evaluates a stress distribution over a virtual polycrystal in terms of discrete harmonic coefficients from diffraction strain data and equilibrium constraints.
- **MechModuli**: Estimates single crystal elastic moduli using diffraction data for grain strains and a virtual sample.
- **MechBands**: Identifies banding within fields based on connected set analyses of either experimental or simulation data

Ancillary Capabilities in MechToolBox:

MechToolBox includes scripts to perform common tasks using the outputs of the core codes and to assist with other pre- and post-processing tasks. Examples include evaluating harmonic expansion coefficients from field data over a virtual polycrystal or orientation data over a fundamental region of Rodrigues space. Scripts for writing VTK and H5 files from any of the core codes are available. Further, scripts that compute grain-average quantities from field data as needed in many of the core codes are included. See the Ancillary Code Overview for more details.

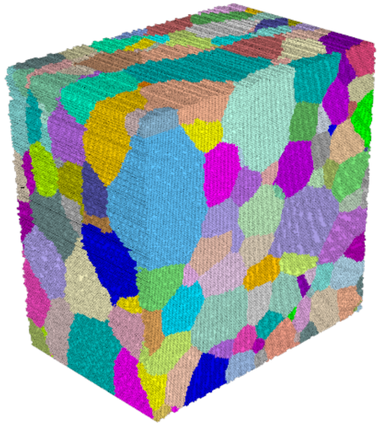
MechToolBox

MechMet

- **MatLab** code for computing mechanical metrics of Virtual Polycrystals
- Utilizes Virtual Polycrystals created by **Neper**
- Solves field equations for elasticity using the finite element method
- From computed displacement fields, computes several mechanical metrics
- Generates a formatted output file for visualization with **Paraview** or **VisIt**

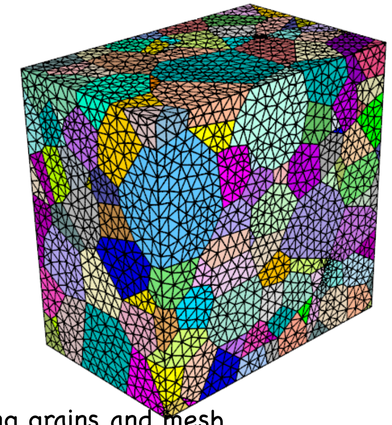
Metrics

- Strain and stress tensors
- Embedded (apparent) directional stiffness
- Isolated (single-crystal) directional stiffness
- Normalized Schmid Factor
- Directional strength-to-stiffness ratio



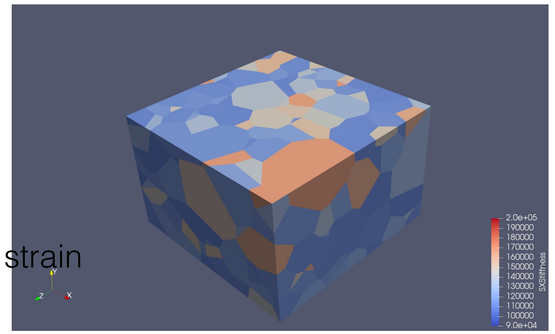
Tesr file image

- HEDM sample with 440 grains (100 pixel threshold)
- Sample data in form of **Neper** .tesr file (D. Pagan)
- S/X properties taken from Ti6Al4V parameters
- Simulations of x, z and x+z direction extension of 0.1% strain
- FE mesh: 81,642 elements and 116,475 nodes

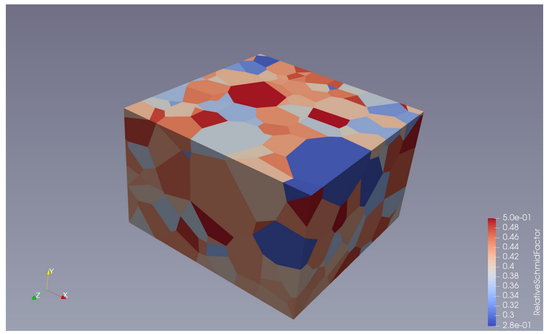


Tessellation showing grains and mesh

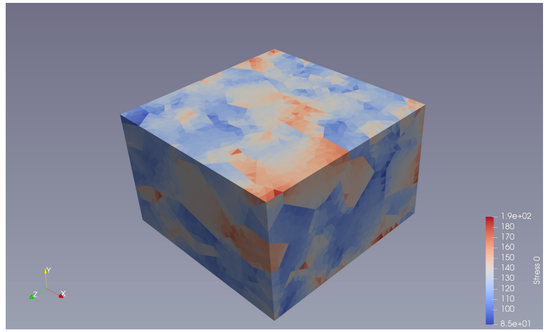
SX-E



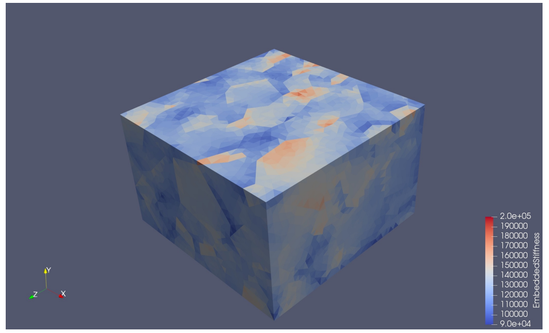
Normalized Schmid Factor



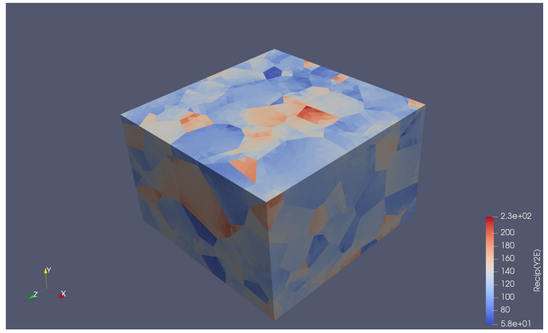
xx component of stress



Embedded E

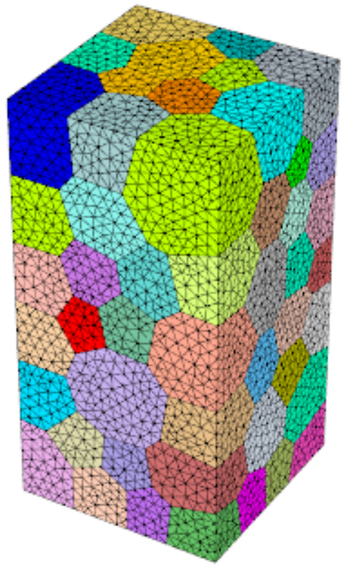


1/Y2E



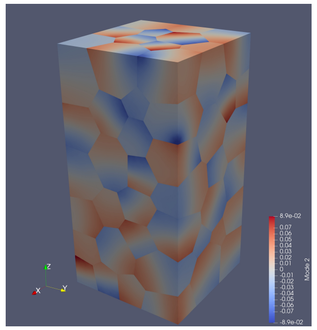
MechMonics

- Computes discrete harmonics for a Virtual Polycrystal
- Bases determined from eigenmodes of the negative Laplacian

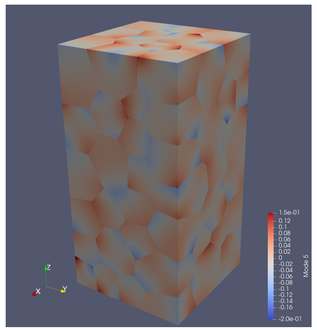


Higher modes display more complex distributions and larger eigenvalues

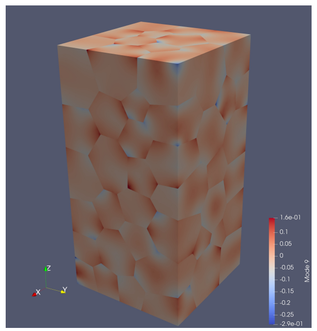
Mode 2

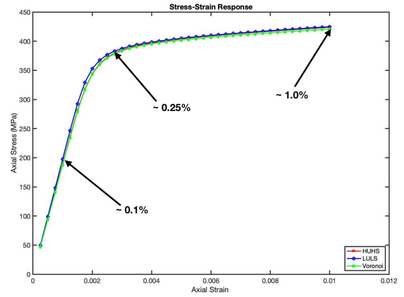


Mode 5

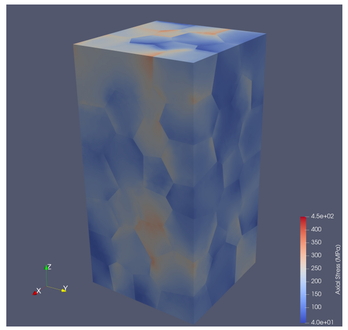


Mode 9

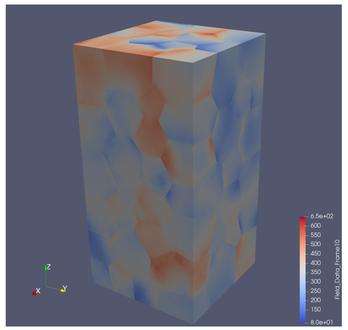




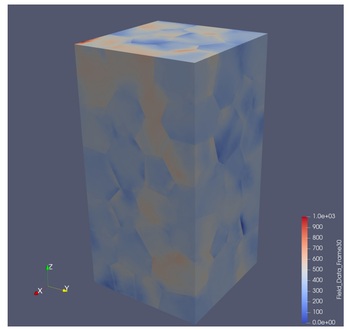
100 grain tensile sample
Mechanical properties for single-phase stainless alloy (fcc)
Sample extended to 1% nominal axial strain
in increments of 0.025% axial strain.
Stress smoothing done grain-by-grain using quad point values



σ_{zz} @ $\epsilon = 0.25\,\%$

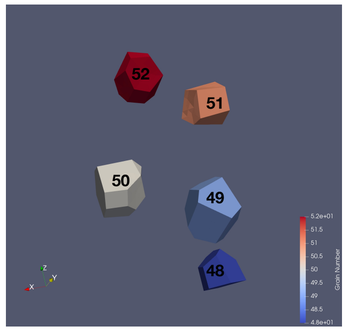


σ_{zz} @ $\epsilon = 0.1\,\%$

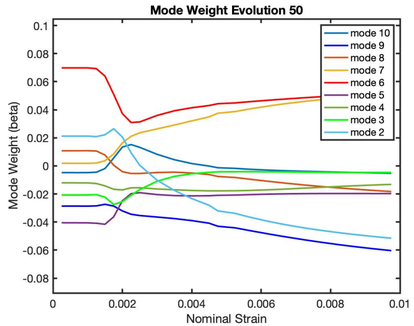


σ_{zz} @ $\epsilon = 1.0\,\%$

Five 'target' grains



Mode weight evolution showing stress redistribution during yielding

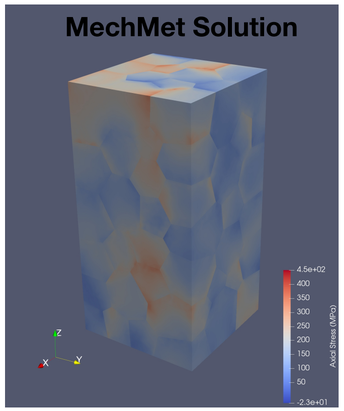


MechReCon

Spatial reconstruction using:

- Harmonic expansions for 6 stress components
- Objective function is a weighted combination of GB traction equilibrium (F^b) and local equilibrium over the grain volume (F^v)
- Equality constraint for HEXD data

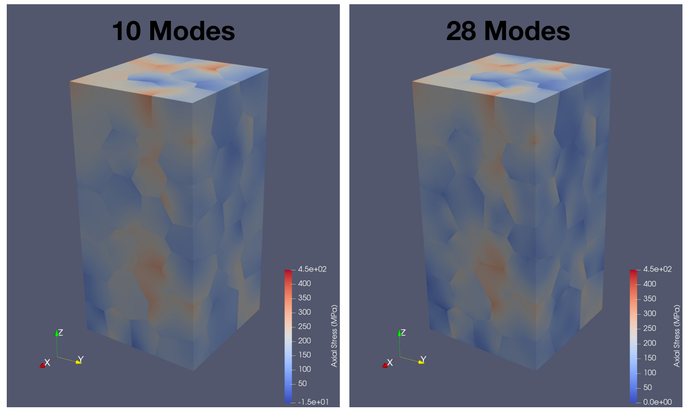
$$F = w^b F^b + w^v F^v$$



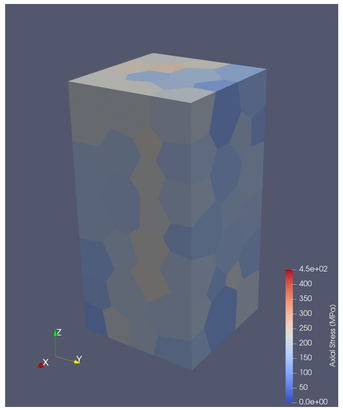
- Z-direction extension of 0.1%
- S/X properties of FCC Al6XN (super-austenistic stainless steel)
- 100 grains

Evaluate harmonic mode coefficients from full stress field

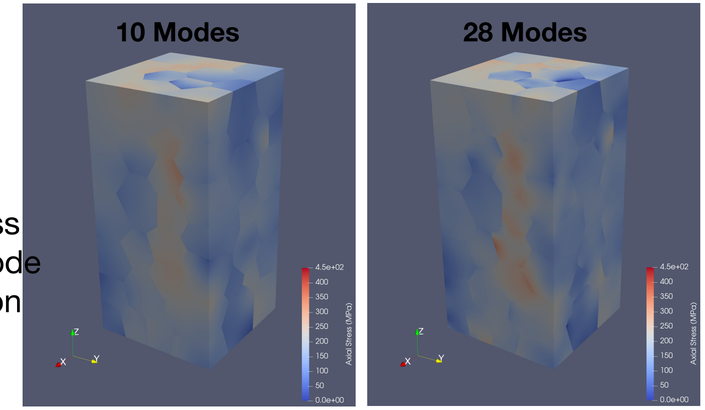
Direct Fit from Original Distribution



Fabricate ‘experimental data’ as average of elemental stresses over each grain



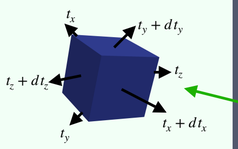
Reconstruct stress using harmonic mode based optimization

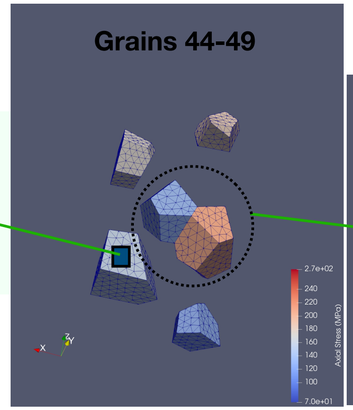


Spatial Reconstruction via Optimization

$$F^v = \{t^v\}^T\{t^v\}$$

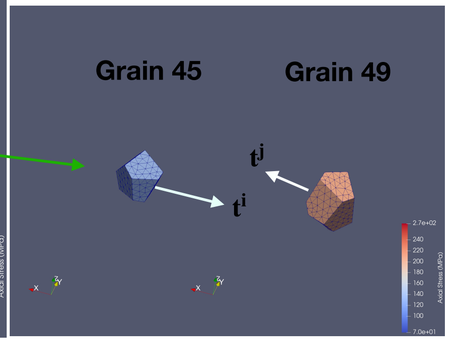
$$\{t^v\} = \{\nabla\sigma\}$$





$$F^b = \{t^b\}^T\{t^b\}$$

$$\{t^b\} = \{t^i + t^j\}$$



MechModuli

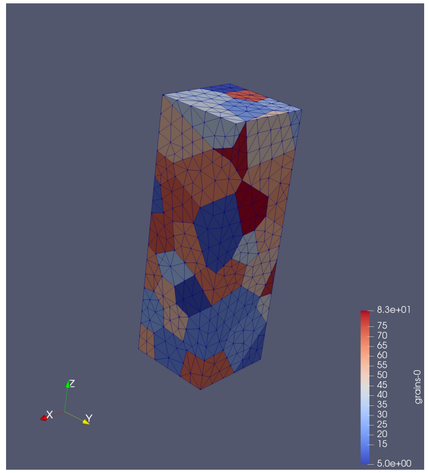
Methodology:

- Optimization based on an objective function defined by the difference between measured and simulated lattice strains:

$$R(\mathbf{S}) = \sum_{Grains} (\varepsilon^{\mathbf{m}} - \varepsilon^{\mathbf{s}}(\mathbf{S}))^{\mathbf{T}} (\varepsilon^{\mathbf{m}} - \varepsilon^{\mathbf{s}}(\mathbf{S}))$$

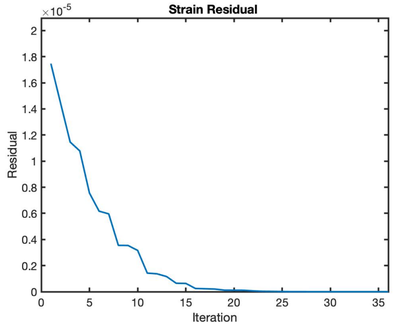
Test Case:

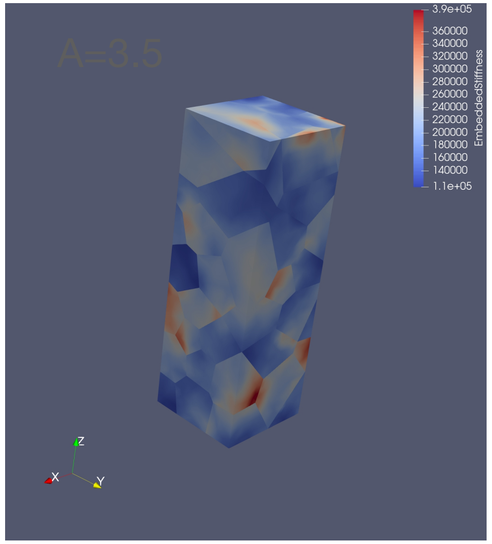
- Synthetic data generated by MechMet
- 83 Grains with 5451 elements and 8337 nodes
- Tensile loading using traction BC
- FCC properties similar to stainless steel



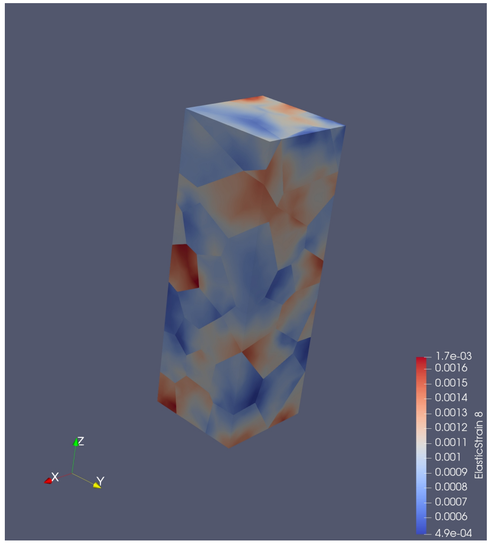
Grains

Strain Residual



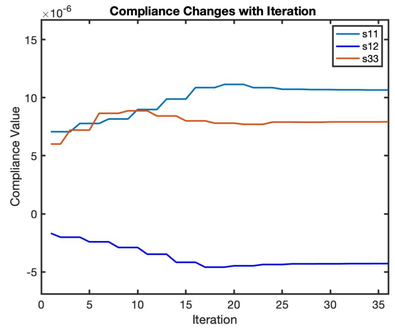


Embedded (apparent) directional elastic modulus



Axial strain at 0.1% extension in z-direction

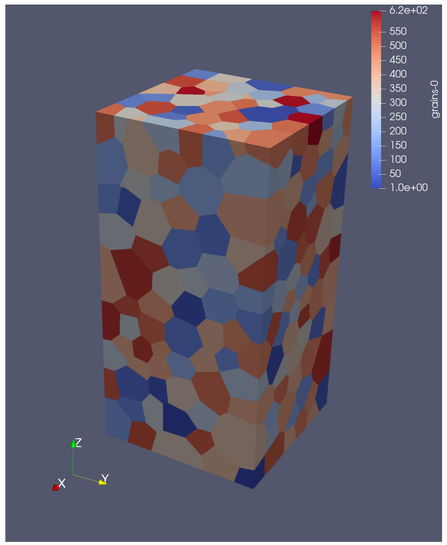
Iteration History - Compliances



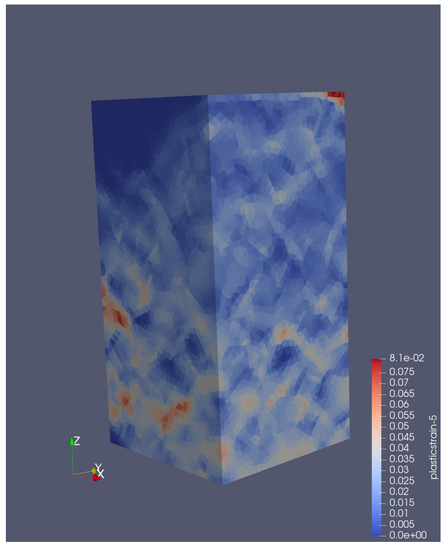
Moduli	C_11	C_12	C_33
A=3.0	204.9	138.0	126.3
A=3.5	204.5	137.6	126.2
A=4.0	204.4	137.8	126.1
MechMet	204.6	137.7	126.2

MechBands

Identify sets of connected elements (graphs) within the mesh of a polycrystal for a specified range of values of a designated field variable.



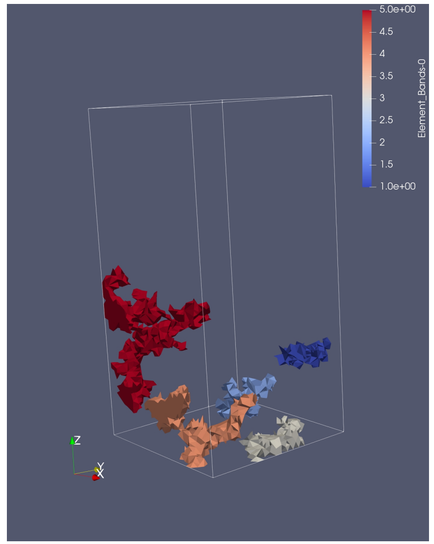
Grain Structure



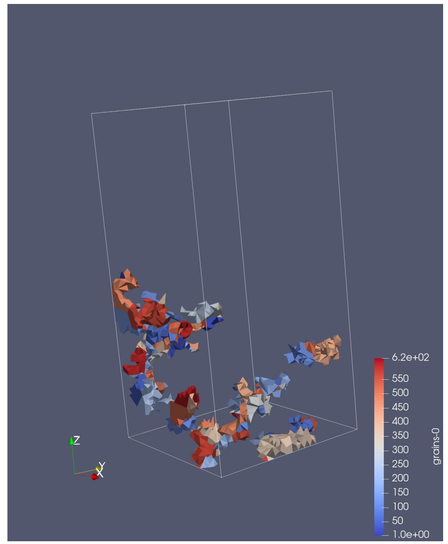
Effective Plastic Strain

Field variable: plastic strain distribution
Deformation state: 2% nominal strain
Solution generated by FEpX.
Polycrystal has 624 grains.
Mesh has ~225,000 elements.

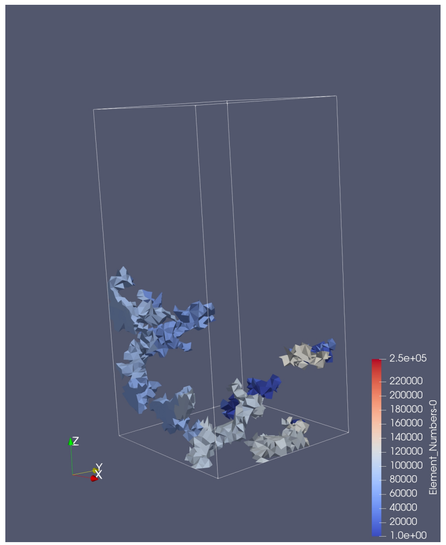
Five Largest Bands



Grain Numbers within Bands



Element Numbers within Bands



Effective Plastic Strain

